

SpaceTravLR CLI convenience cheatsheet

SpaceTravLR_rust (generated)

2026-05-03

spacetravlr, spacetravlr-perturb, and spatial_viewer (crate v2.5.0). Training flags omitted except where they touch prep utilities.
Authoritative list: spacetravlr -help.

Inspect AnnData / files

Flag	Purpose
-peek PATH	Compact summary (shape, obs/var columns, size). Alias -peak. HDF5 metadata for .h5.
-obs COL	With -peek: value_counts for one obs column. With -plot-umap: color UMAP by column.
-h5ad PATH	Main AnnData path; overrides [data].adata_path in config.
-c/-config PATH	spaceship_config.toml overlay.
-infer-species	Print human/mouse from var symbols; requires -h5ad.

Terminal plots

Flag	Purpose
-plot-h5ad	Terminal spatial scatter (obsm spatial); colors by cluster column from config. Needs -h5ad.
-plot-umap [PATH]	Terminal UMAP (obsm['X_umap']); optional path on flag or use -h5ad.
-plot-umap-backend	Values: rust (default) or scanpy (legacy). When UMAP missing, build coords.
-leiden	With -plot-umap: Rust UMAP+Leiden in memory if needed; default coloring by leiden unless -obs.

Rust prep (in-memory or explicit write)

Flag	Purpose
-umap	QC → normalize_total(1e4)+log1p → HVG → PCA → UMAP.
-leiden	Leiden on UMAP graph → obs['leiden'].
-rust-magic	Clusterwise MAGIC → layers['imputed_count'].
-o/-output PATH.h5ad	Write result for modular flags above; omit = memory only.
-rust-process-h5ad	Full Rust pipeline to <stem>_rust_processed.h5ad; requires -h5ad.
-rust-n-top-hvg N	HVG count (default 2000) for -rust-process-h5ad.
-rust-n-neighbors N	KNN/UMAP neighbors (default 15) for -rust-process-h5ad.

Scanpy / uv pipelines (disk outputs)

Flag	Purpose
-process-h5ad	Full Scanpy (QC, UMAP/Leiden) + clusterwise MAGIC → <stem>_processed.h5ad. Alias -process_h5ad.
-impute	MAGIC on layers["normalized_count"] → <stem>_imputed.h5ad (needs cell_type or leiden).
-process-output-dir DIR	Directory for <stem>_*.h5ad from process/impute/rust-process.
-magic-batch-obs COL	MAGIC per (cell type/Leiden) × column; else -condition if set.

Flag	Purpose
<code>-skip-spatial-microns</code>	Skip spatial $\rightarrow \mu\text{m}$ heuristic (<code>-process-h5ad</code>).
<code>-spatial-species</code>	human or mouse prior for spatial scaling.
<code>-spatial-microns-target-um UM</code>	Override median k-NN distance (μm).

CellOracle-style TF priors (exit before train)

Flag	Purpose
<code>-celloracle [PATH]</code>	Bayesian ridge GRN + priors $\rightarrow$ Feather (source, target, cell_type); path optional if <code>-h5ad</code> set.
<code>-celloracle-output PATH</code>	Output Feather path.
<code>-celloracle-output-dir DIR</code>	Default Feather dir + auto preprocess.
<code>-celloracle-layer</code>	Default imputed_count.
<code>-celloracle-skip-preprocess</code>	Do not auto-impute/patch AnnData.
<code>-celloracle-per-cluster</code>	Per cell-type (or <code>-celloracle-obs-key</code>) models.
<code>-celloracle-obs-key</code>	Default cell_type.
<code>-celloracle-species,</code>	GRN species, parquet dir, edge p -threshold, Rayon threads.
<code>-celloracle-network-data-dir,</code>	
<code>-celloracle-p-max, -celloracle-threads</code>	

MALT label transfer (uv + PyTorch on PATH)

Core: `-map-labels -reference REF.h5ad -query QRY.h5ad`.

Flag	Purpose
<code>-map-labels-outdir DIR</code>	Default malt_out.
<code>-g/-map-labels-groupby COL</code>	Label column(s); repeatable or comma-separated in one flag.
<code>-map-labels-output-query NAME</code>	Filename under outdir (default query_labeled.h5ad).
<code>-map-labels-extra-markers GENES</code>	Comma-separated extra genes for dotplots.
<code>-map-labels-expression-mode</code>	auto, counts, or lognorm.
<code>-map-labels-counts-layer LAYER</code>	Raw counts layer.
<code>-map-labels-prefer-raw-counts</code>	Prefer AnnData.raw when resolving counts.
<code>-map-labels-no-leiden</code>	Skip post-MALT Leiden mapping.
<code>-map-labels-reference-gene-list PATH</code>	Gene list file when ref var are placeholders.
<code>-map-labels-spatial</code>	Spatial MALT + betadata priors + neighbor smoothing.
<code>-map-labels-reference-betadata-dir,</code>	SpaceTravLR run dirs (*_betadata.feather).
<code>-map-labels-query-betadata-dir</code>	
<code>-map-labels-benchmark-truth COL</code>	Ground-truth column for metrics.
<code>-map-labels-spatial-genes-per-type N</code>	Default 6.

RCTD (optional crate feature rctd)

`-rctd` with spatial `-h5ad` and `-ref-adata`. See `-help` heading **RCTD**: `-rctd-mode`, `-rctd-output`, `-q-matrices`, `-sigma`, TSV/Q overrides, `-gpu` (with `rctd-wgpu`), reference filters, etc.

run-summary subcommand

`spacetravlr run-summary [-h5ad] [-output-dir] [-c config] [-cluster-key] [-layer] [-run-id] [-manifest] [-betadata-pattern]` — HTML report without training.

spacetravlr-perturb

Flag	Purpose
<code>-run-toml PATH</code>	<code>spacetravlr_run_repro.toml</code> .
<code>-export / -out PATH</code>	Batch: write simulated Feather (needs <code>-gene</code>).
<code>-gene NAME</code>	Single-gene batch.
<code>-desired-expr, -n-propagation</code>	Perturbation parameters.
<code>-cells-csv, -cells-csv-column</code>	Cell scope from CSV columns of obs_names.

Flag	Purpose
<code>-batch-toml PATH</code>	Multi single-gene jobs (not with <code>-gene / -export / -cells-*</code>).
<code>-batch-parallelism N</code>	Override file parallelism.
<code>-config / -perturb-toml PATH</code>	Overlay TOML (optional positional config).
<code>-v/-verbose</code>	Timings / extra logging.

spatial_viewer (feature spatial-viewer)

Flag	Purpose
<code>-h5ad, -layer, -cluster-annot</code>	Dataset defaults (<code>imputed_count, cell_type</code>).
<code>-bind, -port</code>	HTTP server (default <code>127.0.0.1:8080</code>).
<code>-static-dir</code>	Built web assets (default <code>web/spatial_viewer/dist</code>).
<code>-run-toml</code>	Enables perturbation + betadata from run.
<code>-perturb-overlay</code>	TOML merged into run config.
<code>-network-dir</code>	Parquet GRN directory.
<code>-allow-cors</code>	Permissive CORS for MCP iframe.
<code>-perturb-cache-dir, -foyer-grn-memory-mb,</code> <code>-foyer-grn-disk-mb, -foyer-grid-memory-mb,</code> <code>-foyer-grid-disk-mb</code>	Foyer cache path and tier sizes (MiB).

Training-adjacent

Flag	Purpose
<code>-skip-auto-adata-prep</code>	Do not auto Scanpy/impute when layers or <code>cell_type</code> missing.
<code>-plain, -verbose, -demo</code>	Logs / fake dashboard.
<code>-update, -update-version TAG</code>	Self-update (feature <code>self-update</code>).

Compute env (training / some backends)

`SPACETRAVLR_FORCE_CPU=1` or `SPACETRAVLR_DISABLE_WGPU=1` forces CPU ndarray path instead of WebGPU.